		Total time = $t_1 + t_2 = 4.224$ s
6	**D**	electric power × efficiency = gain in gravitational PE per unit time so, electrical power = $\frac{400 \times 9.81 \times 1200}{120} \div 0.80 = 49$ kW
7	**A**	By conservation of energy, both coins begin with the same gravitational potential energy,